

SAJAN CHAKKUMKATTUPARAMBIL RAJU

Cybersecurity Engineering | Detection Engineering and SOC | Threat Intelligence | MITRE
ATT&CK | Breach and Attack Simulation

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PROFESSIONAL SUMMARY

M.Sc. Cybersecurity graduate (Vistula University, Warsaw) specialising in detection engineering, SOC platform development, breach and attack simulation, and threat intelligence. Built and lab-tested a four-platform defensive ecosystem from first principles: SentinelForge (real-time SOC detection and automated response), ControlProbe (MITRE ATT&CK coverage scoring against a live Wazuh SIEM), DeceptionGrid (distributed honeypot network capturing real attacker TTPs and exporting STIX 2.1 CTI bundles), and a quantum-resistant cryptography system (M.Sc. thesis, ML-KEM/Kyber, SHA3-512, Ed25519). Worked as SOC Analyst Intern at BSH Technologies, producing documented security findings and multi-stakeholder reports. Completed structured cybersecurity simulations with Datacom, Deloitte, and EY. M.Sc. thesis on post-quantum cryptographic systems. Legally authorised to work full-time in Poland.

CORE COMPETENCIES

Detection Engineering: Suricata IDS rule development, Wazuh SIEM custom rule authoring, MITRE ATT&CK technique mapping, alert triage, deduplication engine design, coverage gap analysis, Sigma rule format

Threat Intelligence: STIX 2.1 CTI bundle production (threat-actor, attack-pattern, indicator, observed-data), OpenCTI and MISP ingestion, AbuseIPDB IP enrichment, campaign timeline reconstruction (DeceptionGrid)

SOC Operations: Alert triage and root cause analysis, IOC extraction, phishing and BEC investigation, log correlation across Linux auth and IDS streams, iptables automated block response, incident documentation

Security Tooling: Suricata IDS, Wazuh SIEM, Splunk, ELK Stack, Nessus, Wireshark, TCPDump, Nmap, Metasploit, Burp Suite, Microsoft Sentinel, AbuseIPDB, VirusTotal, PESTudio

Breach and Attack Simulation (BAS): 15 MITRE ATT&CK technique execution across 9 tactics, Wazuh REST API polling for detection validation, A-D coverage scoring, gap analysis reporting (ControlProbe)

Honeypot and Deception Technology: Distributed SSH, HTTP, FTP honeypot design, Paramiko-based SSH server, attacker profiling, multi-protocol campaign detection, attacker sophistication scoring

Cryptography and Secure Systems: Post-quantum cryptography (ML-KEM/Kyber, AES-256-GCM), Ed25519 signatures, SHA3-512 and HMAC-SHA3-512 integrity verification, hybrid encryption system design (M.Sc. thesis)

Platform Engineering: Python (FastAPI, Flask, Pydantic v2, Paramiko, cryptography), Docker, Git, SQLite, REST API design and integration, Chart.js dashboards, Bash scripting

PROFESSIONAL EXPERIENCE

SOC Analyst Intern

October 2025 – January 2026

BSH Technologies Pvt. Ltd., Bhubaneswar, India (Remote from Warsaw, Poland)

Faculty-approved placement, Vistula University | Supervisor-proposed grade: 4.5 / 5.0

- Investigated security alerts across Linux and network environments, performing triage and root cause analysis on authentication anomalies, suspicious outbound connections, and IDS signature matches;

documented each case with affected assets, observed indicators, severity assessment, and recommended remediation steps in a format ready for senior review.

- Identified three recurring threat patterns within two months by correlating multi-source log data; submitted findings as candidate detection improvements, two of which were adopted into the team's monitoring workflow.
- Analysed phishing and BEC-style email samples by inspecting headers, SPF and DKIM authentication records, embedded URLs, and attachment metadata; extracted IOCs and produced structured investigation reports without additional triage required by the next reviewer.
- Built standardised investigation note templates covering observed behaviour, affected systems, severity context, and remediation guidance; templates were adopted into the team's shared process within the first month.
- Contributed draft response procedures for three recurring incident types and supported vulnerability assessment activities, adding to the team's operational knowledge base.

Cybersecurity Operations Training Programme

2024 – 2025

Red Team Hacker Academy, Intensive SOC and Infrastructure Training (six months)

- Completed 20+ structured investigation exercises using Splunk, Wazuh, and Snort: alert triage, IOC identification, adversarial technique recognition, and written reporting for both technical and non-technical audiences.
- Conducted packet-level traffic analysis using Wireshark and TCPDump across DNS, HTTP, SMTP, and TLS sessions, identifying patterns consistent with reconnaissance, credential abuse, and command-and-control communication.
- Applied Python scripting to automate log parsing, event correlation, and report generation, reducing manual overhead on repetitive investigation steps.

PROFESSIONAL SIMULATIONS — FORAGE (BIG 4 AND TOP-TIER FIRMS)

Cyber Security Operations Analyst | **Datacom**

June 2026

Forage Virtual Simulation | Security Analysis, Risk Assessment, OSINT, Incident Investigation, Risk Management

- Investigated a simulated cyberattack from initial indicator through to root cause: ran OSINT on threat actor infrastructure, mapped attack vectors, and wrote a client-ready security report with ranked remediation recommendations in the format Datacom uses in live client engagements.
- Produced a structured risk assessment translating technical findings into business impact terms, matching Datacom's active client output format.

Cyber Security Analyst | **Deloitte Australia**

June 2026

Forage Virtual Simulation | Log Analysis, Web Security, Python, Data Analysis, Breach Investigation

- Analysed web server activity logs to trace the source of a simulated data breach and identify suspicious user sessions, following Deloitte's documented incident investigation methodology.
- Used Python to surface access anomalies not visible in raw log output, mirroring the scripted analysis approach used in active breach response engagements.

Technology Risk Analyst | **EY (Ernst & Young)**

June 2026

Forage Virtual Simulation | IT Risk Analysis, Internal Controls, Risk Assessment, Professional Skepticism

- Assessed IT risk scenarios using EY's Technology Risk methodology: reviewed internal controls, identified control gaps, and applied professional skepticism — directly replicating the fieldwork applied in technology risk and cyber maturity engagements.

FLAGSHIP CYBERSECURITY PROJECTS

SentinelForge: Real-Time SOC Detection and Response Platform github.com/sajancr3/sentinelforge-soc-platform 2025

Python, Suricata IDS, Flask, MITRE ATT&CK, iptables, Bash | Two-VM lab: attacker Ubuntu vs. Debian SOC

- Built a real-time SOC platform from scratch: ingests Suricata IDS alerts and Linux auth logs, runs a custom detection engine that maps every alert to a MITRE ATT&CK technique, deduplicates events with a correlation window, scores cumulative risk per source IP, and displays a live event feed on a Flask dashboard.
- Engineered automated response: when cumulative risk exceeds threshold (70/100), the platform generates an iptables DROP rule and applies it to the attacker IP with no analyst input — reducing mean time to contain.
- Lab-tested against real attack scenarios using a two-VM environment: Nmap network recon (847 Suricata events collapsed to one correlated incident, blocked in 4 minutes 12 seconds), SSH brute-force, and lateral movement simulation.

ControlProbe: Breach and Attack Simulation / SIEM Coverage Validation 2025 – 2026 github.com/sajancr3/controlprobe

Python, FastAPI, Wazuh REST API, MITRE ATT&CK, Docker, AbuseIPDB, Chart.js

- Executed 15 real MITRE ATT&CK techniques across 9 tactics (Initial Access, Execution, Persistence, Privilege Escalation, Defence Evasion, Credential Access, Discovery, Lateral Movement, Exfiltration), polled Wazuh SIEM via REST API for each technique, and scored detection coverage A-D.
- Key finding from live run: Wazuh out-of-box rules scored C (61/100). Lateral movement (T1021.002 SMB) and exfiltration (T1048) generated zero alerts — primary gap areas requiring custom Sigma/KQL rules.
- Produced structured gap analysis reports identifying which controls failed and why, with remediation recommendations — the same deliverable a cybersecurity auditor presents in a SIEM maturity review. Commercial equivalents (Cymulate, AttackIQ) cost \$50,000–\$200,000 per year.

DeceptionGrid: Distributed Honeypot Network and Threat Intelligence Platform 2025 – 2026 github.com/sajancr3/deception-grid

Python, Paramiko, STIX 2.1, Docker, AbuseIPDB, FastAPI, SQLite | Debian ARM64 lab node

- Designed and deployed a distributed honeypot network (SSH, HTTP, FTP protocols) in Docker: each service captures credentials attempted, commands executed, and session metadata, feeding a central SQLite database.
- Lab result: 6,500+ attack events captured in four hours from a Hydra-based SSH brute-force campaign. Three distinct source IPs across two ASNs identified as coordinated (not opportunistic), peaking at 847 attempts per minute.
- Attacker profiling pipeline: AbuseIPDB enrichment (two of three IPs at 95%+ abuse confidence), attack timeline reconstruction, credential pattern analysis, and attacker sophistication scoring.
- CTI export: STIX 2.1 bundles containing threat-actor, attack-pattern, indicator, and observed-data objects, ingested successfully into OpenCTI and compatible with MISP and QRadar feeds.

Quantum-Resistant Cryptography Computer System 2025 – 2026 *M.Sc. Thesis, Vistula University | Debian Linux, ML-KEM/Kyber, AES-256-GCM, Ed25519, SHA3-512*

- Designed and implemented a hybrid post-quantum encryption system combining ML-KEM/Kyber key encapsulation with AES-256-GCM symmetric encryption, providing quantum resistance for key exchange while retaining symmetric performance for data encryption.
- The Ed25519 and SHA3-512 components from this research were directly applied to the TrustShield IAM evidence vault and the DORA platform cryptographic sealing layer, establishing a provable chain of custody for audit findings.

M.Sc. Computer Science: Cybersecurity and Reliability of IT and Industrial Systems

Graduated June 2026

Vistula University (Akademia Finansów i Biznesu Vistula), Warsaw, Poland

Relevant modules: Digital Forensics and Malware Analysis, Security of Computer Systems, Computer Networks and Protocols, Risk Analysis, Linux and Ubuntu Operating Systems, Cloud Technologies, Wireless and Mobile Security.

Thesis: Design and Implementation of a Quantum-Resistant Cryptography Computer System (ML-KEM/Kyber, AES-256-GCM, Ed25519, SHA3-512, Debian Linux).

B.Sc. Computer Applications

University of Calicut, India

CERTIFICATIONS AND PROFESSIONAL CREDENTIALS

CICSA (Certified Information and Cybersecurity Analyst) | Red Team Hacker Academy, 2024 | SIEM monitoring, IDS alert triage, incident response, SOC operations

ANZ Cybersecurity Virtual Experience Programme | Forage | Phishing and BEC-style email analysis, header inspection, IOC extraction, banking-sector incident classification

ISO 27701 Lead Auditor | Mastermind, 2026 | PIMS audit, GDPR controls, privacy data governance

Datacom Cyber Security Operations | Forage, 2026 | Cyberattack investigation, risk assessment, OSINT, client-ready reporting

Deloitte Cyber Security Analyst | Forage, 2026 | Log analysis, breach investigation, Python-assisted data analysis

EY Technology Risk Virtual Programme | Forage, 2026 | IT risk assessment, internal controls, professional skepticism

WORK ELIGIBILITY AND LANGUAGES

Work authorisation: Legally authorised to work full-time in Poland. EU work authorisation documents available for HR on request. **Languages:** English: C1 Professional | Polish: Conversational, actively advancing |

Malayalam: Native | Hindi: Fluent